

BIL 401 / BIL 501 Big Data and Distributed Data Processing Project Descriptions

Research Track (at most and preferably 2 students)

This option involves

- Reading multiple papers in specific problem space
- Proposing a new idea that might possibly improve the scalability or performance metrics
- Implementing the idea(e.g. modifying an existing open source framework)
- Running tests, Collecting perf metrics and comparing against the baseline
- Writing a research paper which can ideally be submitted to a conference or workshop
- Although desired results may fail to perform better than the baseline, Observed Level of effort will be considered for grading the project outputs.

Analytics(Applied) Track (at most and preferably 3 students)

This track involves

- Finding a rich dataset or set of datasets that can be processed to infer useful conclusions. (possibly synthetic)
- Installing and setting up an Apache Spark based data pipeline for processing the dataset(s)
- Designing and implementing your algorithms in Spark (free to use any API, i.e., RDD, dataframe, dataset etc. any language pyspark, scala, R or Java)
- Basic visualization of the output metrics
- Writing a paper in an academic paper format
- In this track Spark is your friend. Go research MLLib, GraphX or any third party library
- Note: Your dataset does not have to be too big as long as the project idea makes sense and you use spark to do the processing

Systems Track (at most and preferably 3 students)

This track involves:

- Focusing on infrastructure engineering and orchestration by integrating at least three or four distinct open-source distributed technologies (e.g., Spark, Kafka, Cassandra, Citus).
- Deploying these technologies using Docker and Docker Compose to simulate a distributed environment locally.
- Simulating and testing the scalability, connectivity, and fault-tolerance of the integrated system.
- Writing a technical report detailing the architecture, setup, and performance evaluations.

- *Note:* The scope should be feasible to execute and test locally on students' laptops rather than using large-scale cloud clusters.

Machine Learning Data Engineering Track (at most and preferably 3 students)

This track involves:

- Focusing on scalable data preparation and ML pipelines.
- Designing pipelines for training data preparation or processing features for low-latency inference.
- Emphasizing feature engineering and integrating data management layers, such as feature stores, alongside model behavior.
- Implementing and testing these pipelines using Spark or other distributed frameworks.
- Writing a technical report detailing the pipeline architecture and engineering performance metrics.
- *Note:* Focus on smaller, high-quality, and manageable datasets that fit local execution constraints.

BIL501 Projects needs to be from the Research track or high quality applied science ideas

BIL401 Projects are recommended to be from the Analytics track.

Checkpoints (upload on uzak.etu.edu.tr)

-End of Week 4 16.06.2026)

Submit

- One paragraph description of the project
- Source(URL or scraping method) of the dataset
- Description of the dataset (domain, columns, rough estimate of the size... etc.)
- Platforms, frameworks, tools, cloud systems, hardware you are planning to use

-End of Week 7 (TBD, 23:59)

Submit a Status Report describing

- State of data collection (ideally should be done by this time)
- Status of the platforms and systems to use
- installation problems, know-how, demo runs
- Paper with abstract, related work, proposed implementation sections.

- Final Report (TBD, 23:59)

- Submit All material(code, data source if possible) project implementation/platforms details and paper
- Will schedule a **demo day** that each group will have 20 mins to present(demo) their work

If you think you have very good ideas that do not fit the format here, talk to me very **early**, I might approve it.

If you realize (mid-semester) that your project proposal is not feasible, talk to me for a plan-B.

[illegible]

DATASETS

- <https://www.resmiiistatistik.gov.tr/>
- <https://datasetsearch.research.google.com/>
- <https://www.kdnuggets.com/datasets/index.html>
- <https://www.kaggle.com/datasets>
- <https://www.ee.columbia.edu/~cylin/course/bigdata/getdatasetinfo.html>
- <https://www.icpsr.umich.edu/web/pages/>
- <https://dataverse.harvard.edu/>
- <https://networkrepository.com/soc.php>
- <https://evds2.tcmb.gov.tr/index.php?evds/serieMarket>
- <https://paperswithcode.com/datasets>
- <https://webscope.sandbox.yahoo.com/>
- <http://snap.stanford.edu/data/index.html>
- <https://ieee-dataport.org/datasets>
- <https://courses.cs.washington.edu/courses/cse547/21sp/data.html>
- <https://developers.google.com/freebase>
- https://www.wikidata.org/wiki/Wikidata:Main_Page
- <https://datasf.org/opendata/>
- <https://webdatacommons.org/productcorpus/index.html>

About Web scraping:

<https://medium.com/analytics-vidhya/scrapy-vs-selenium-vs-beautiful-soup-for-web-scraping-24008b6c87b8>

Project Inspirations

Distributed Graph Centrality Metrics
Distributed Graph algos (Dijkstra.. etc.)
Common item matching across datasets where a join key is not present (Cimri.com, akakçe.com)

Linear Programming gibi optimizasyon problemlerini (sparkda) dağıtık modelleyip çalıştırmak
Distributed Postgresql scalability tests - Citus
Spark - Kafka Connection, scaling
Spark - Cassandra Connection, scaling
Implementing an anomaly Detection Algo in Spark MLLib , dbscan, isolation forest, etc..
Implementing a Recommendation Systems Algo in Spark MLLib
Training/Testing an ML algo in Spark MLLib using GPU
Distributed Columnar Data Generator that supports tunable column correlations
Optimizing Speculative execution in MapReduce
Predicate push down strategies, ideas
Leveraging Network Encoding in Big Data shuffles
Inspire from this list